

Thesis Figures — Standalone Preview

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This is not the thesis itself. Mohammed's Track A edits happen in `Chapters/chapter*.tex`; this preview is a sandbox to confirm the visual look before / during embedding.

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Chapter 1

Chapter 2 — Literature Review & Background

Figure 2.1 *RAG system architecture*

Planned section: §2.1

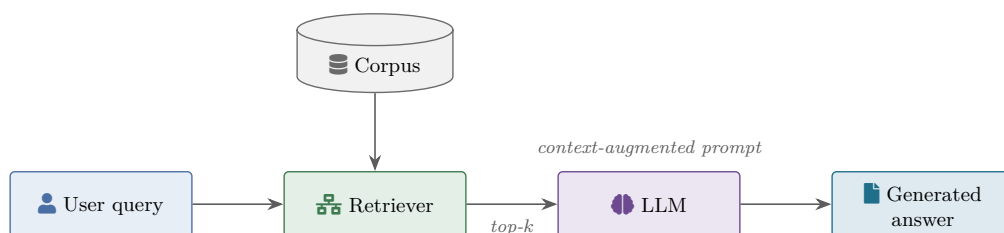


Figure 1.1: High-level architecture of a Retrieval-Augmented Generation (RAG) system. A user query is sent to a retriever that consults the corpus and returns the top- k passages, which are passed alongside the query to an LLM to produce a context-augmented answer.

Example in-body cross-reference:

“The Retrieval-Augmented Generation framework (Figure 1.1) couples a retriever with a generator, with the query enhancement layer of this thesis sitting between the user query and the retriever.”

Table 2.1 *Reviewed query-enhancement papers (13 entries)*

Planned section: §2.4

Table 1.1: Query-enhancement papers reviewed for this thesis, with method family, evaluation dataset, and headline result.

Paper	Year	Family	Method	Dataset
Query2Doc (Wang et al.)	2023	generation	LLM pseudo-document concatenation	MS MARCO + BEIR
HyDE (Gao et al.)	2022	generation	LLM hypothetical document	TREC-DL / BEIR
GAR (Mao et al.)	2021	generation	LLM generates answer/context for query rewriting	open-domain QA
GRF (Mackie et al.)	2023	generation	Generative Relevance Feedback for dense	TREC-DL
CSQE (Lei et al.)	2024	generation	Corpus-Steered Query2Doc	TREC-DL
MuGI (Zhang et al.)	2024	generation	Multi-Text Generation Integration	BEIR
ThinkQE (Lei et al.)	2025	generation	Iterative chain-of-thought expansion	TREC-DL
Exp4Fuse (Liu et al.)	2025	fusion	Original + expanded query via modified RRF	multiple
DAPR (Wang et al.)	2024	generation	Dense Approach via Pre-trained Retrievers	academic IR
AQE (Yang et al.)	2025	generation	Aligned Query Expansion via LLM alignment	academic IR
KAR (Xia et al.)	2025	generation	Knowledge-Aware Query Expansion	textual + relational retrieval
Song & Zheng (survey)	2024	survey	Query optimisation taxonomy (4 ops)	–
QE survey (arXiv 2509.07794)	2025	survey	Comprehensive QE survey in LLM era	–

Example in-body cross-reference:

“Table 1.1 summarises the query-enhancement literature, grouped by whether the technique is lexical (PRF, RM3, BM25 expansion) or LLM-based generative (Query2Doc, HyDE, CSQE, MuGI, ThinkQE, Exp4Fuse).”

Table 2.2 *LLM models used in this thesis (11 entries)*

Planned section: §2.5

Table 1.2: Open-source LLMs evaluated for Arabic query enhancement, ordered by parameter count.

Model	Family	Params (B)	Multilingual	Arabic-specialised	Quantisation
SILMA Kashif-2B	Silma AI	2	No	Yes (RAG)	FP16
Qwen 2.5 3B	Alibaba	3	Yes (29 langs)	No	FP16
Falcon-H1-3B	TII	3	Yes	Yes	BF16
Qwen3-4B	Alibaba	4	Yes (119 langs)	No	FP16
Gemma 3 4B	Google	4	Yes	No	FP16/BF16
Qwen 2.5-7B	Alibaba	7	Yes (29 langs)	No	FP16
ALLaM-7B (DROPPED)	NCAI	7	Yes	Yes	FP16
Jais-2-8B	G42	8	Yes	Yes	BF16
Qwen3-8B	Alibaba	8	Yes (119 langs)	No	FP16
Aya Expanse 8B	Cohere	8	Yes (101 langs)	No	4-bit NF4
GPT-OSS 20B (DROPPED)	OpenAI	20 (MoE)	Yes	No	4-bit NF4

Example in-body cross-reference:

“The eleven models in Table 1.2 span 2 B–20 B parameters, with both multilingual generalists (Qwen 2.5, Qwen3, Gemma 3, Aya Expanse) and Arabic-specialised models (SILMA Kashif, Falcon-H1, Jais-2, ALLaM).”

Chapter 2

Chapter 3 — Methodology

Figure 3.1 *End-to-end thesis pipeline*

Planned section: §3.1

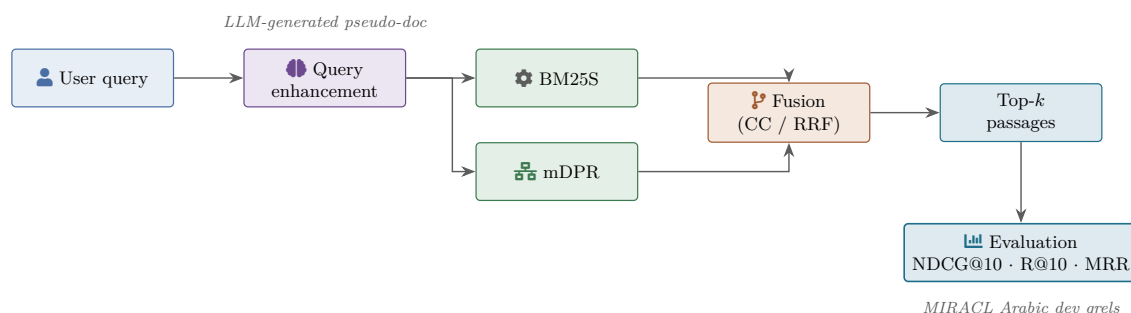


Figure 2.1: The thesis evaluation pipeline. A user query passes through the query-enhancement layer, after which BM25S and mDPR retrieve independently; their ranked lists are then fused (CC or RRF) and the top- k passages are scored against the MIRACL Arabic dev qrels.

Example in-body cross-reference:

“Figure 2.1 shows the overall evaluation pipeline used throughout this thesis; subsequent sections describe each stage in detail.”

Figure 3.3 *BM25S indexing flow*

Planned section: §3.4

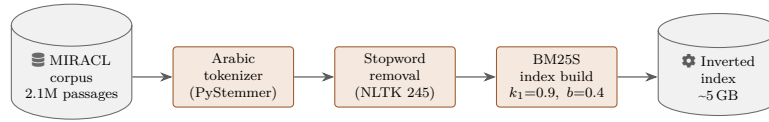


Figure 2.2: BM25S indexing pipeline applied to the MIRACL Arabic corpus. Passages are tokenised with PyStemmer, stopwords-removed using 245 Arabic stopwords, then indexed under BM25S with $k_1=0.9$ and $b=0.4$.

Example in-body cross-reference:

“The sparse retrieval pipeline (Figure 2.2) uses BM25S, a pure-Python BM25 implementation, with Arabic-aware tokenisation and stopwords removal.”

Figure 3.4 *mDPR encoding flow*

Planned section: §3.5

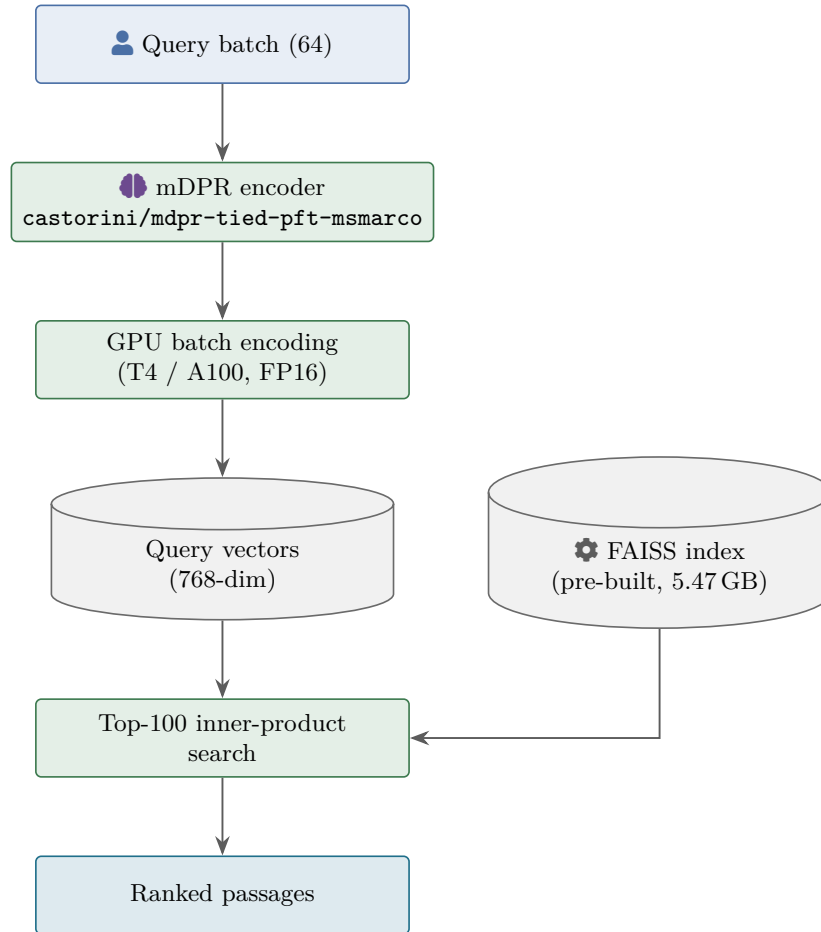


Figure 2.3: Dense retrieval pipeline using multilingual DPR. Queries are encoded in batches on a single A100 GPU under FP16; the resulting 768-dimensional vectors are searched against a pre-built FAISS index of the corpus.

Example in-body cross-reference:

“Dense retrieval (Figure 2.3) is implemented via the multilingual DPR encoder `castorini/mdpr-tied-p` applied to the pre-built MIRACL FAISS index.”

Figure 3.5 *Query2Doc generation*

Planned section: §3.6

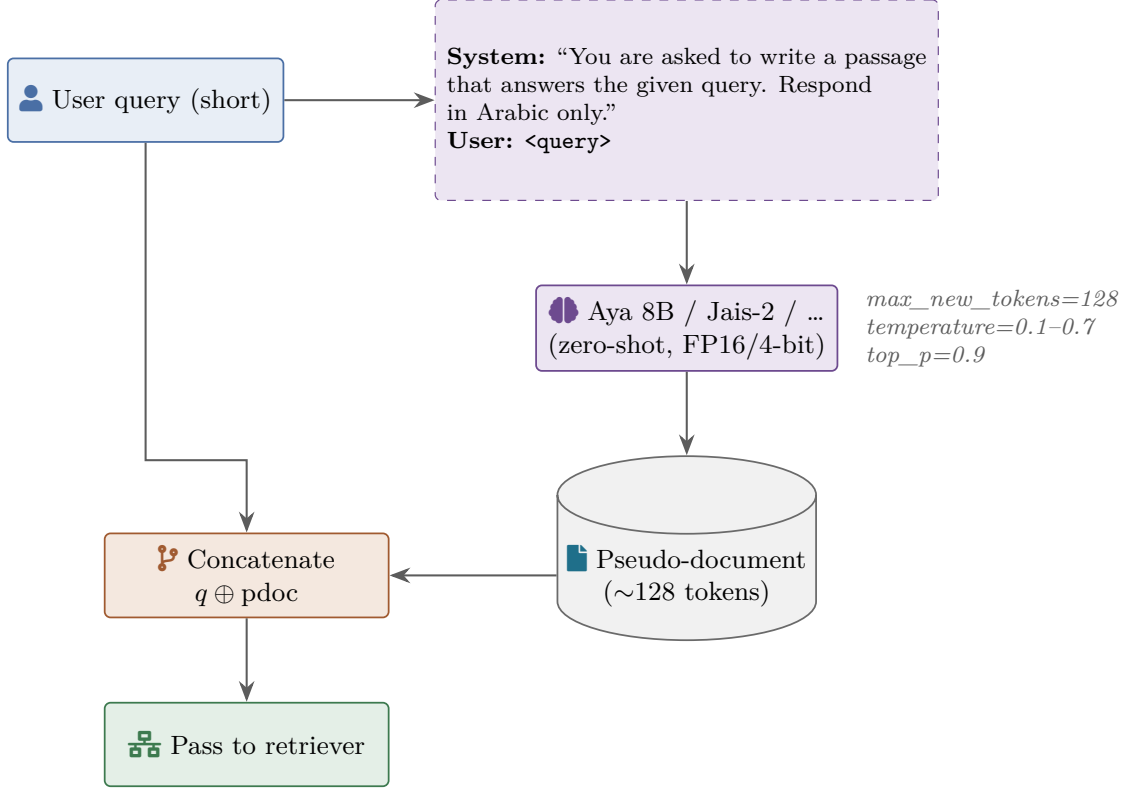


Figure 2.4: Query2Doc generation: the original query is wrapped in a fixed prompt template, an open-source LLM generates a short pseudo-document conditioned on the query, and the pseudo-document is concatenated with the query before being passed to the retriever.

Example in-body cross-reference:

“The Query2Doc pipeline (Figure 2.4) follows the original Wang et al. (2023) formulation, with the LLM and generation parameters varied per experiment.”

Figure 3.7 Hybrid fusion (Convex Combination and RRF)

Planned section: §3.7

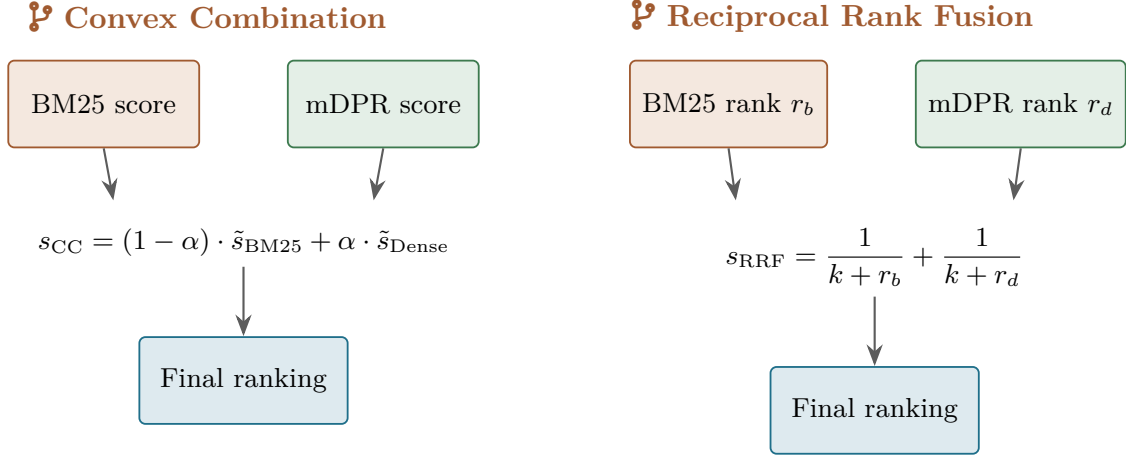


Figure 2.5: Two retriever-fusion methods evaluated in this thesis. Convex Combination (left) computes a weighted sum of normalised retrieval scores; Reciprocal Rank Fusion (right) sums the reciprocal of each retriever’s rank for every document.

Example in-body cross-reference:

“Two complementary fusion strategies (Figure 2.5) are evaluated: CC requires score normalisation, while RRF is parameter-light and operates on ranks alone.”

Figure 3.8 *CSQE pipeline (three-stage)*

Planned section: §3.8

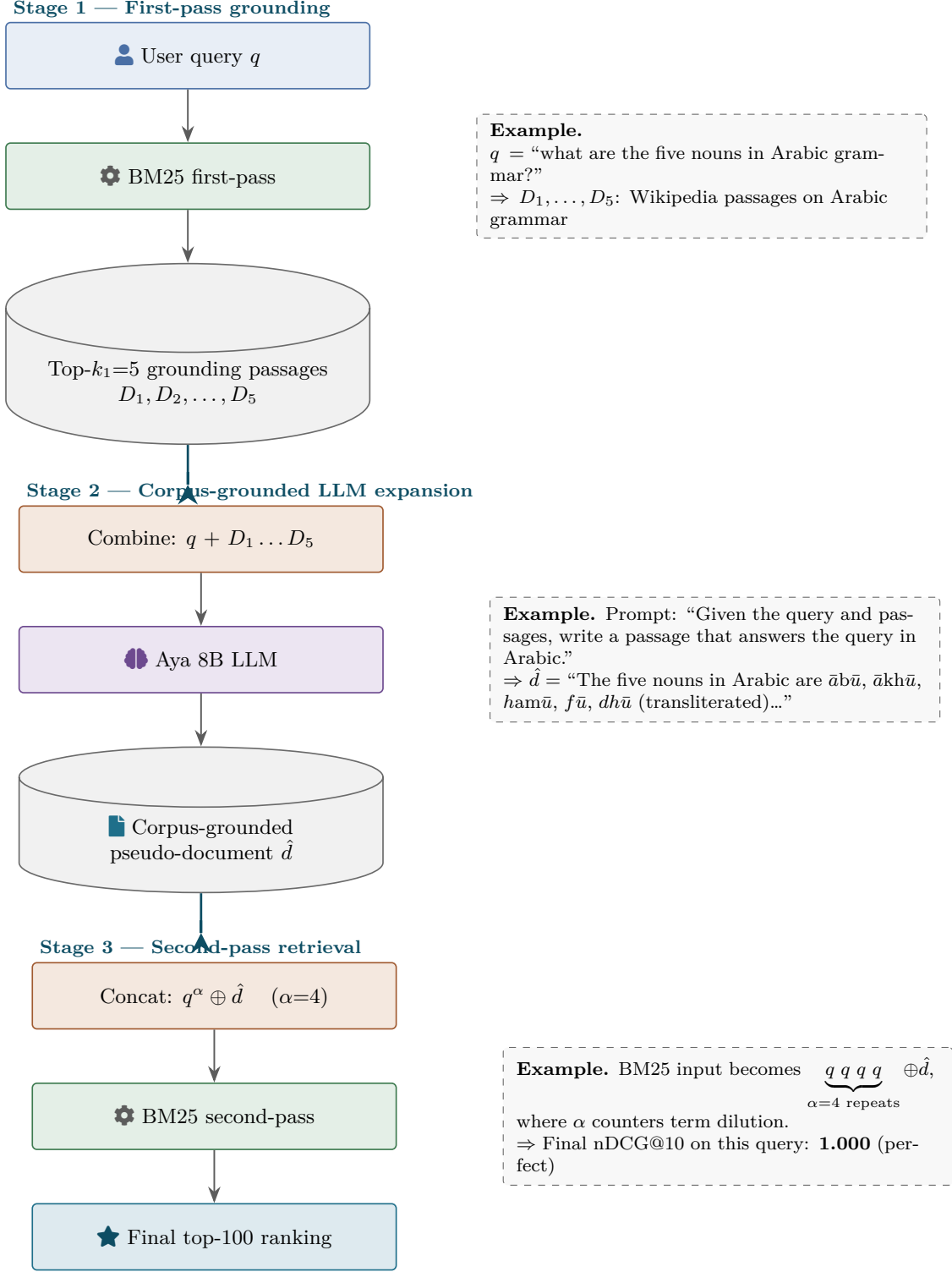


Figure 2.6: Corpus-Steered Query Expansion (CSQE) pipeline. Stage 1: BM25 retrieves a first-pass top- k_1 context. Stage 2: corpus context plus the original query are fed to the Aya LLM, which generates a corpus-grounded pseudo-document. Stage 3: the query (repeated α times) is concatenated with the pseudo-document and fed to a BM25 second pass.

Example in-body cross-reference:

“The corpus-steered variant (Figure 2.6) replaces the blind LLM expansion of Figure 2.4 with one grounded in the corpus, mitigating the entity-hallucination failures observed in the blind setting.”

Figure 3.9 *Best system architecture (BM25-only-expanded RRF $k = 20$)*

Planned section: §3.8.3 / §4.7

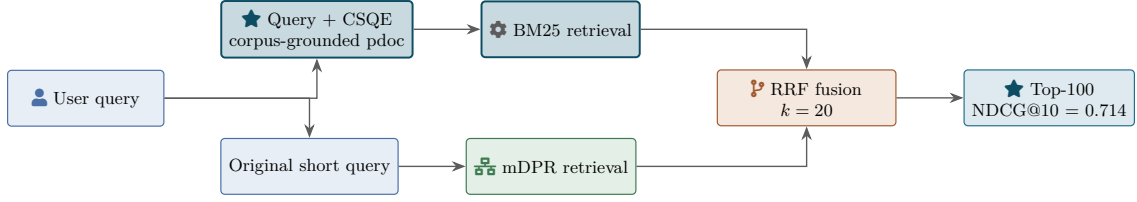


Figure 2.7: Architecture of the thesis’s best-performing system. CSQE-expanded queries are fed to BM25 while the original short queries are fed to mDPR; the two ranked lists are fused with Reciprocal Rank Fusion ($k = 20$), yielding the headline nDCG@10 of 0.714.

Example in-body cross-reference:

“The best-performing configuration (Figure 2.7) applies expansion asymmetrically: only the sparse retriever receives the CSQE-grounded query.”

Table 3.1 *Per-model hardware configuration (9 rows)*

Planned section: §3.6

Table 2.1: Hardware and precision configuration used for each LLM evaluated, plus the per-model batch size and total wall-clock runtime on the MIRACL dev set.

Model	GPU	Precision	Batch	Runtime (min)	Notes
SILMA Kashif-2B	A100 40GB	FP16	16	15–20	–
Qwen 2.5 3B	A100 40GB	FP16	8	~40	–
Falcon-H1-3B	A100 40GB	BF16	1	~60	hybrid Mamba; batching broken
Qwen3-4B	A100 40GB	FP16	32	~25	enable_thinking=False
Gemma 3 4B	A100 40GB	FP16/BF16	16	~30	numerical-stability fixes (BF16)
Qwen 2.5-7B	A100 40GB	FP16	16	~35	–
Jais-2-8B	A100 40GB	BF16	8	~50	token_type_ids removed; BF16 mandatory (Squared-ReLU o
Qwen3-8B	A100 40GB	FP16	16	~40	thinking tags stripped
Aya Expanse 8B	A100 40GB	4-bit NF4	8	~50	–

Example in-body cross-reference:

“All experiments ran on a single A100 40 GB GPU under the configurations listed in Table 2.1.”

Table 3.2 *Per-model generation hyperparameters (9 rows)*

Planned section: §3.6

Table 2.2: Pseudo-document generation hyperparameters per model. Temperature, top- p and max_new_tokens were tuned per model where the model card specified recommended values; default otherwise.

Model	Temp.	Top-p	Max tokens	Special
SILMA Kashif-2B	0.1000	0.9000	128	temp 0.1 chosen empirically over 0.7 (+2.5%)
Qwen 2.5 3B	0.7000	0.9000	128	original Query2Doc default
Falcon-H1-3B	0.1000	0.9000	128	model-recommended (>0.1 drops performance)
Qwen3-4B	0.7000	0.9000	128	enable_thinking=False; do_sample=True (greedy gives infinite repetitions)
Gemma 3 4B	0.1000	0.9000	128	–
Qwen 2.5-7B	0.1000	0.9000	128	–
Jais-2-8B	0.7000	0.9000	128	token_type_ids not passed
Qwen3-8B	0.1000	0.9000	128	thinking tags stripped at decode time
Aya Expanse 8B	0.1000	0.9000	128	–

Example in-body cross-reference:

“Generation hyperparameters per model are summarised in Table 2.2; the largest model-specific deviations were Falcon-H1’s recommended temperature of 0.1 and Qwen3’s mandatory `enable_thinking=False`.”

Chapter 3

Chapter 4 — Results & Discussion

Table 4.1 *Baselines: mDPR vs BM25*

Planned section: §4.1

Table 3.1: Baseline retrieval performance on the MIRACL Arabic dev set (2,896 queries). All metrics rounded to three decimals.

Retriever	Recall@10	Recall@100	NDCG@10	MRR
mDPR (Dense)	0.616	0.841	NaN	NaN
BM25S (Sparse)	0.596	0.858	NaN	NaN

Example in-body cross-reference:

“As Table 3.1 shows, mDPR and BM25 are complementary: BM25 has higher recall at 100 but lower precision in the top 10.”

Figure 4.1 *Per-query $nDCG@10$ distribution (mDPR vs BM25)*

Planned section: §4.1

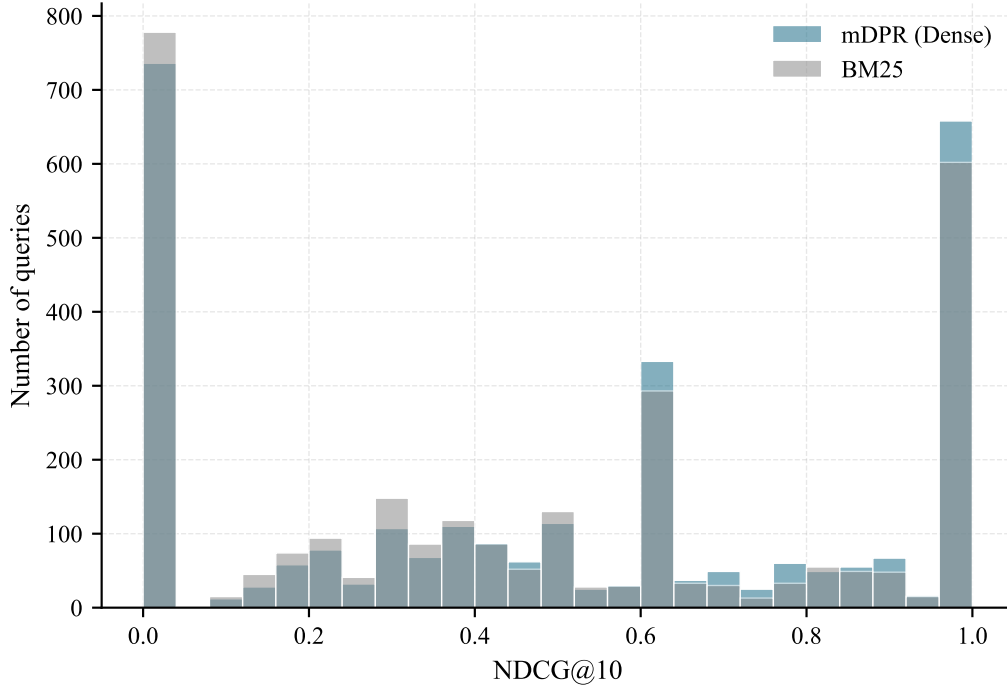


Figure 3.1: Per-query $nDCG@10$ distribution for the two baseline retrievers on the MIR-ACL Arabic dev set. Both retrievers exhibit a bimodal distribution: a large mass of queries near zero (poorly served) and another mass near one (perfectly served), with relatively few in the middle range.

Figure 4.2 *Failure cliff* — 33.9% of queries score $nDCG@10 < 0.3$

Planned section: §4.2.1

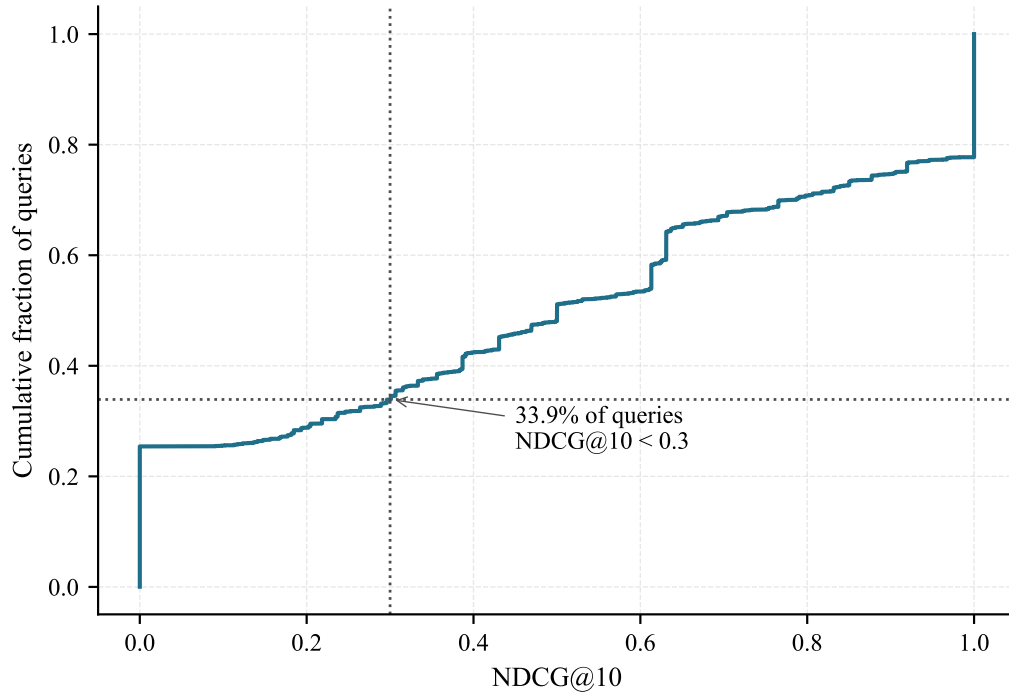


Figure 3.2: Cumulative distribution of per-query $nDCG@10$ on the mDPR baseline. The marked threshold at 0.3 separates 33.9% of queries on which the baseline performs poorly from the remainder.

Figure 4.3 $nDCG@10$ by query length (1–3 / 4–8 / 9+ words) — *v2 violin*

Planned section: §4.2.2

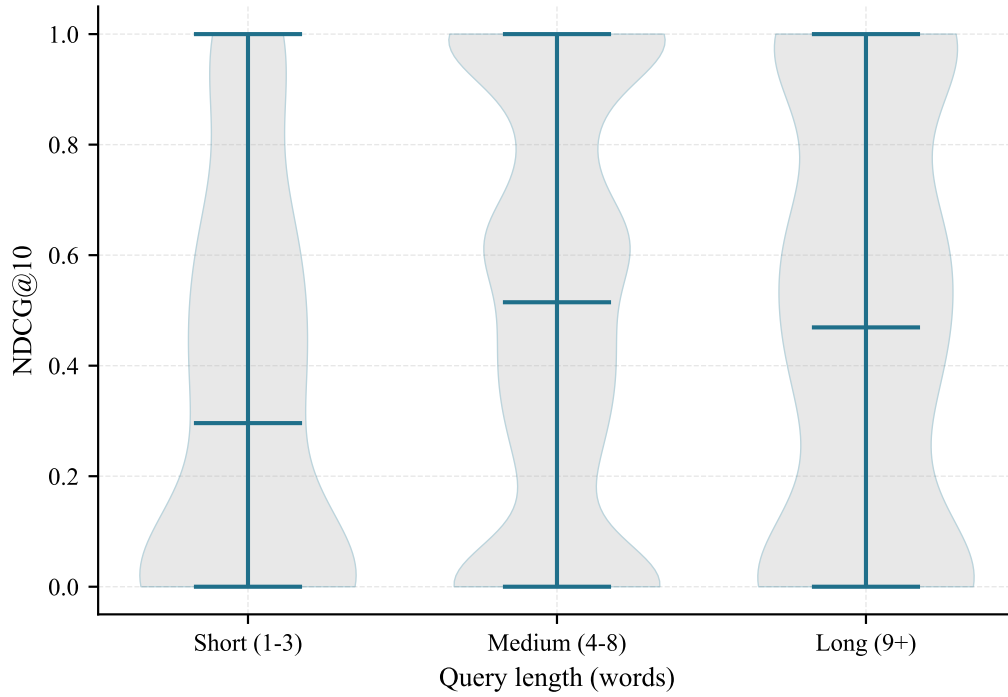


Figure 3.3: Distribution of mDPR $nDCG@10$ within each query-length bucket. Short queries (1–3 words) achieve the lowest mean $nDCG@10$ (0.345) while medium queries (4–8 words) achieve the highest (0.511); the relationship is non-monotonic, not a linear trend.

Figure 4.4 *Recall@k curve, $k = 1 \dots 100$*

Planned section: §4.2.3

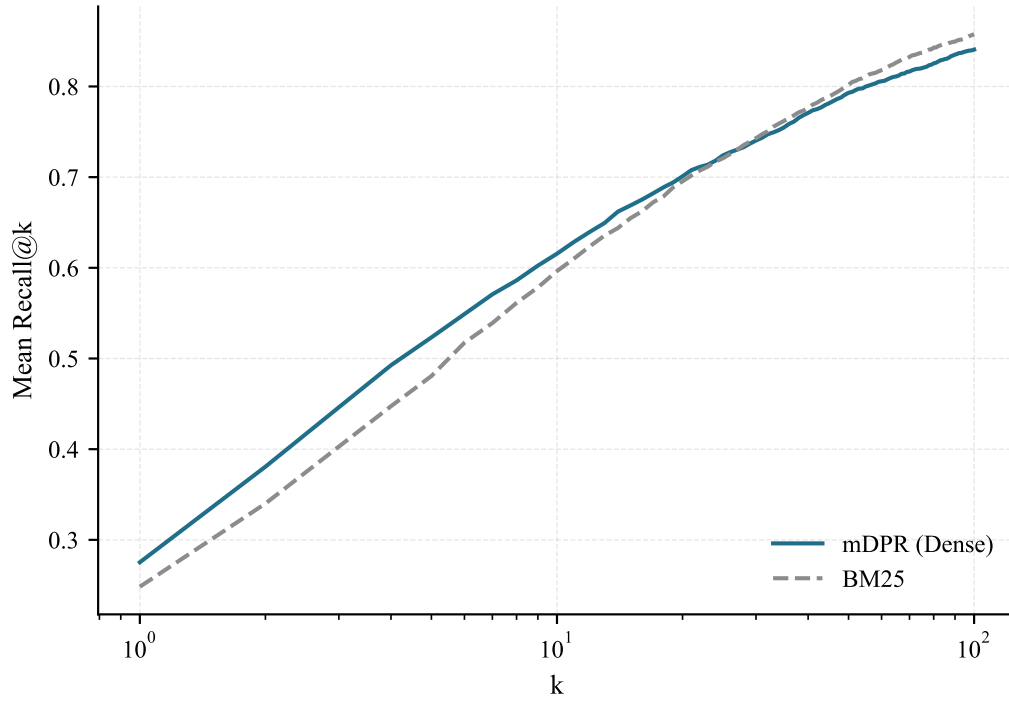


Figure 3.4: Mean Recall@ k for mDPR and BM25 on the MIRACL Arabic dev set, k on a log axis. Approximately 10% of relevant passages remain unrecovered even at depth 100, motivating hybrid retrieval and corpus-steered expansion.

Table 4.2 *Dense-retrieval results across 10 LLMs*

Planned section: §4.3

Table 3.2: Query2Doc dense-retrieval results across ten LLMs, ordered by nDCG@10. Aya Expanse 8B leads at 0.616.

Model	Params (B)	nDCG@10	R@10	R@100	MRR	Δ vs baseline	Notes
Aya Expanse 8B	8.0000	0.6164	0.7256	0.9001	0.6493	0.1171	blind QE (4-bit NF4); BEST OVERALL
Jais-2-8B	8.0000	0.6018	0.7161	0.8981	0.6356	0.1025	–
Qwen3-8B	8.0000	0.5958	0.7119	0.8877	0.6278	0.0965	thinking tags stripped
Aya 8B CSQE	8.0000	0.5915	0.7073	0.8816	0.6225	0.0922	CSQE variant
Qwen 2.5-7B	7.0000	0.5813	0.6952	0.8800	0.6134	0.0820	–
Qwen3-4B	4.0000	0.5691	0.6824	0.8726	0.6015	0.0698	–
Qwen 2.5 3B	3.0000	0.5435	0.6608	0.8594	0.5742	0.0442	–
Gemma 3 4B	4.0000	0.5391	0.6443	0.8477	0.5761	0.0398	–
Falcon-H1-3B	3.0000	0.5359	0.6484	0.8531	0.5681	0.0366	–
SILMA Kashif-2B	2.0000	0.5177	0.6289	0.8353	0.5508	0.0184	temp 0.1 (chosen over 0.7)
mDPR (no QE)	–	0.4993	0.6156	0.8407	0.5328	0.0000	baseline
ALLaM-7B	7.0000	0.2550	0.3335	0.5465	0.2708	-0.2443	DROPPED (tokenizer bug)

Figure 4.5 v1 *Sorted $nDCG@10$ across 10 LLMs (vertical)*

Planned section: §4.3 — also include v3 grouped per Osman

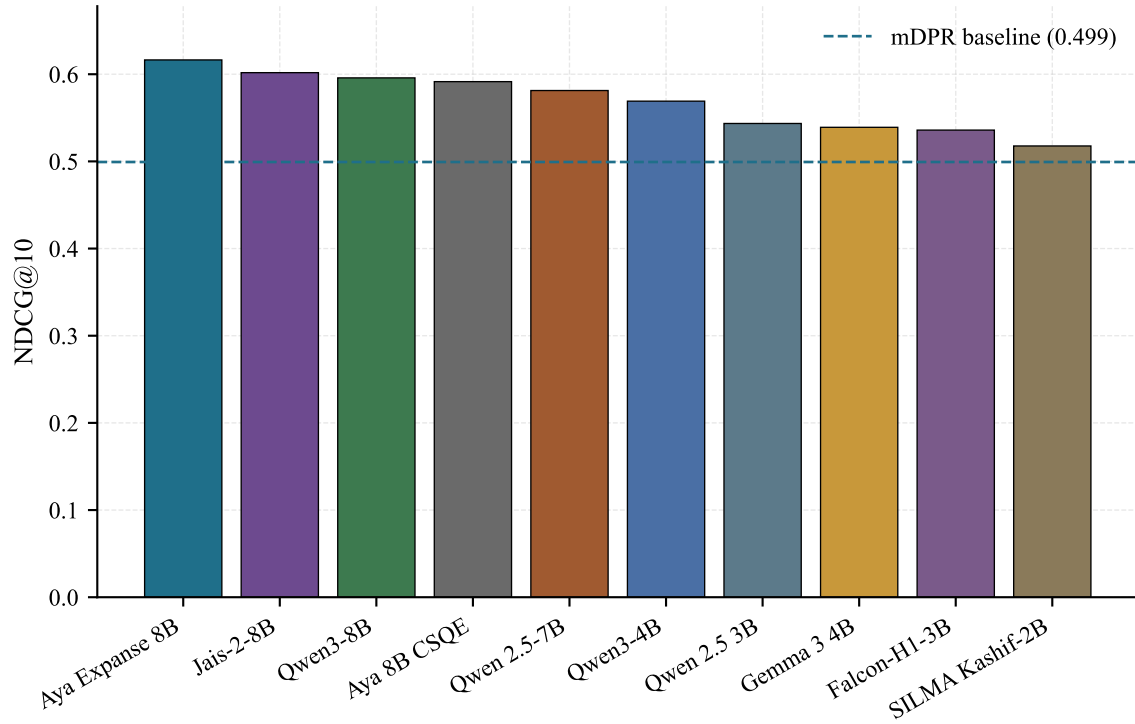


Figure 3.5: $nDCG@10$ of each evaluated LLM on the Query2Doc dense pipeline, sorted descending. The dashed reference line marks the mDPR baseline (0.499); each model uses its consistent thesis colour, identical to Figure 3.8.

Figure 4.5 v3 *Multi-metric grouped variant*

Planned section: §4.3 (companion to Fig 4.5)

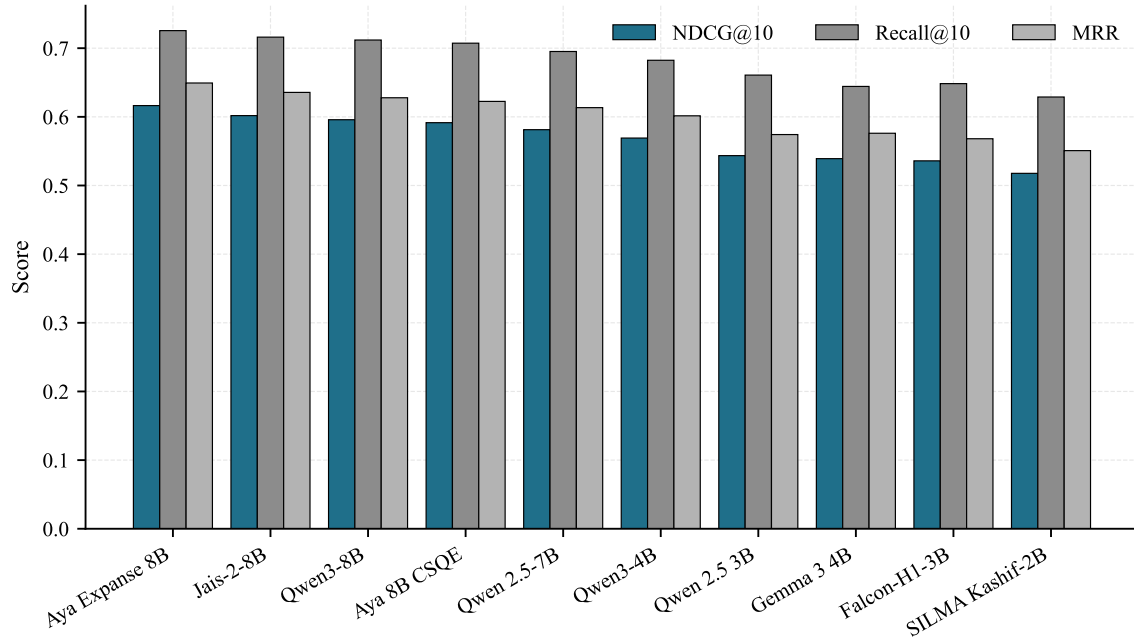


Figure 3.6: Grouped per-model bars for nDCG@10, Recall@10 and MRR on the Query2Doc dense pipeline.

Figure 4.6 v2 *Model size vs $\Delta nDCG@10$ (labelled + trendline)*

Planned section: §4.3

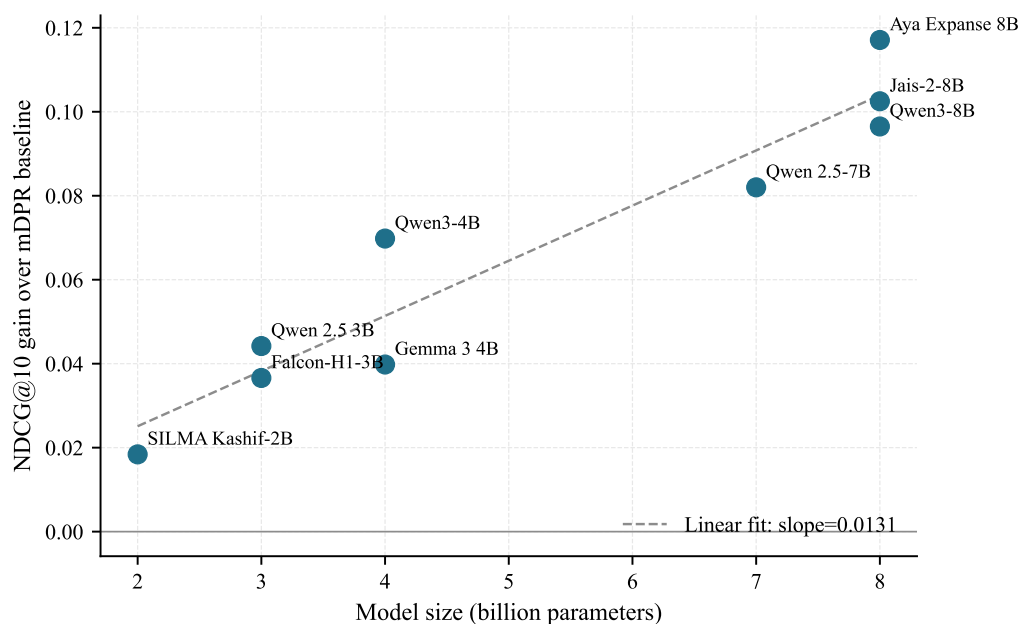


Figure 3.7: Relationship between model parameter count (billions) and gain over the mDPR baseline. A weak positive trend is visible; the Qwen-family subset (3B–8B) corroborates the size-quality correlation within a single architecture.

Table 4.3 *Per-model BM25 best configurations (9 rows)*

Planned section: §4.4 / §4.6

Table 3.3: Best BM25 query-repetition configuration per model. All nine models exceed the BM25 baseline (0.462) at their optimum repetition count.

Model	Params (B)	$n=1$ nDCG@10	$n=1$ R@10	$n=1$ R@100	$n=1$ MRR	Best config	Best nDCG@10	Δ
BM25 baseline (no QE)	–	0.4621	0.5964	0.8577	0.4836	$n=1$	0.4621	
SILMA 2B	2.0000	0.4277	0.5550	0.8115	0.4485	$n=5$	0.4832	
Falcon-H1-3B	3.0000	0.4038	–	–	–	$n=10$	0.5113	
Qwen 2.5-3B	3.0000	0.4090	–	–	–	$n=5$	0.5185	
Qwen3-4B	4.0000	0.4145	–	–	–	$n=7$	0.5244	
Gemma 3 4B	4.0000	0.3447	0.4532	0.7182	0.3718	$n=7$	0.5277	
Qwen 2.5-7B	7.0000	0.4682	0.6040	0.8646	0.4905	$n=5$	0.5358	
Qwen3-8B	8.0000	0.4459	0.5806	0.8499	0.4702	$n=7$	0.5328	
Jais-2-8B	8.0000	0.5122	–	–	–	$\text{beta}=2$	0.5731	
Aya Expanse 8B	8.0000	0.5046	0.6284	0.8734	0.5377	$\text{beta}=2$	0.5855	

Figure 4.7 *BM25 query-repetition sweep across 9 models*

Planned section: §4.6

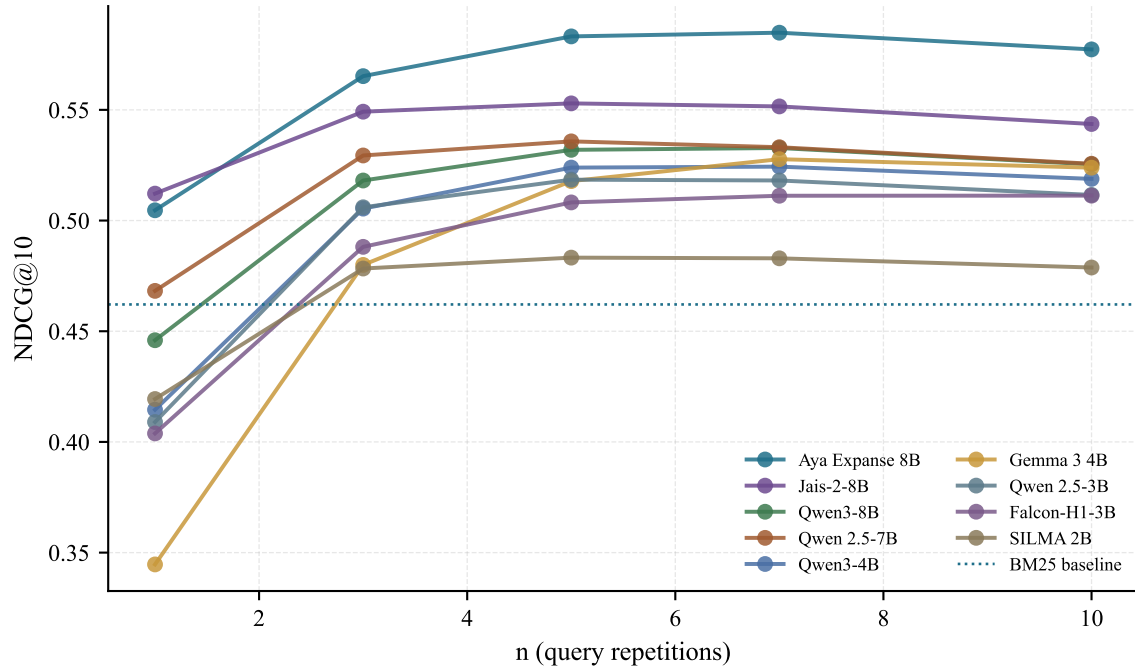


Figure 3.8: nDCG@10 on the BM25 retriever as a function of query-repetition count n for each LLM-generated pseudo-document. Six of nine models start below the BM25 baseline at $n=1$; all nine recover or exceed it at their optimum n . Colours match Figure 3.5.

Figure 4.8 *Dense gain vs BM25 gain per model*

Planned section: §4.4 / §4.6

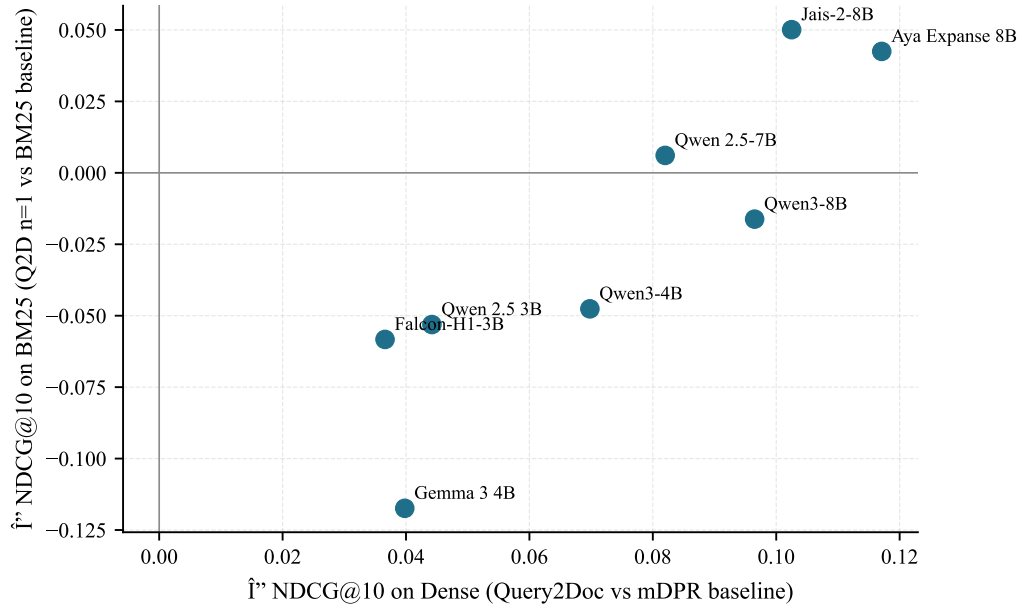


Figure 3.9: Per-model Query2Doc gain on the Dense retriever versus the BM25 retriever. Only Aya Expanse 8B and Jais-2 8B sit in the both-positive quadrant; most other models help Dense but degrade BM25 at $n=1$.

Table 4.4 *Hybrid baselines (no QE)*

Planned section: §4.7

Table 3.4: Hybrid retrieval baselines without query enhancement: BM25 alone, mDPR alone, and the two fusion variants.

Method	nDCG@10	Recall@10	Recall@100	MRR
BM25 alone	0.462	0.596	0.858	0.484
mDPR alone	0.499	0.616	0.841	0.533
Hybrid CC ($\alpha=0.5$)	0.627	0.748	0.946	0.658
Hybrid RRF (k=20)	0.627	0.760	0.947	0.652

Figure 4.9 v2 *Hybrid CC α sweep — all four metrics*

Planned section: §4.7

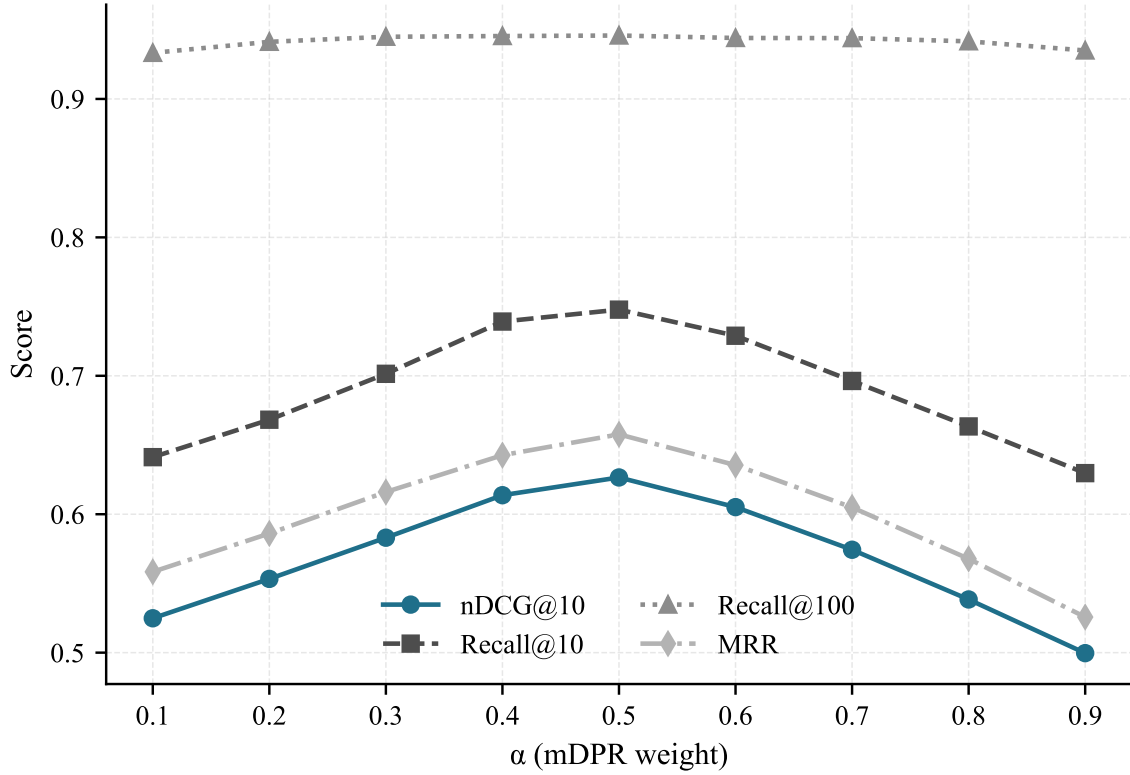


Figure 3.10: Convex-combination α sweep for the no-QE hybrid baseline, showing nDCG@10, Recall@{10,100} and MRR. nDCG@10 peaks near $\alpha = 0.5$; Recall@100 is largely insensitive to α .

Table 4.5 *CSQE component ablation*

Planned section: §4.8

Table 3.5: Ablation of the CSQE pseudo-document, comparing corpus-only, blind-only, and the 2-corpus + 2-blind combination on BM25 alone and within the best hybrid configuration.

System	nDCG@10	Recall@10	Recall@100	MRR
BM25 alone (no QE)	0.462	0.596	0.858	0.484
Hybrid RRF k=20 (no QE)	0.627	0.760	0.947	0.652
Config A RRF best (Exp 2.1)	0.714	0.836	0.973	0.736
013d: Blind-only BM25 (0c+4b alpha=4)	0.575	0.709	0.920	0.603
013c: Corpus-only BM25 (4c+0b alpha=4)	0.538	0.646	0.879	0.565
013: CSQE BM25 (2c+2b alpha=4)	0.616	0.745	0.942	0.638
013d + Dense RRF (Config A)	0.708	0.829	0.961	0.732
013c + Dense RRF (Config A)	0.662	0.790	0.951	0.681
013 + Dense RRF (Config A)	0.714	0.836	0.973	0.736

Table 4.6 *CSQE vs blind QE on BM25 + Dense*

Planned section: §4.8

Table 3.6: Comparison of CSQE against blind Query2Doc using Aya Expanse 8B as the generator, evaluated on the BM25 and Dense retrievers in isolation.

Variant	BM25 nDCG@10	Dense nDCG@10
No QE (baseline)	0.4621	0.4993
Aya 8B blind QE	0.5855	0.6164
Aya 8B CSQE	0.6157	0.5915

Table 4.7 *Configs (BM25-only-expanded / Dense-only-expanded / Both-expanded)*

Planned section: §4.9

Table 3.7: Performance of the three retriever-assignment configurations across CC and RRF fusion, with the no-QE hybrid baseline and BM25/CSQE / Dense/CSQE running alone for reference.

Method	nDCG@10	Recall@10	Recall@100	MRR
BM25 alone	0.462	0.596	0.858	0.484
mDPR alone	0.499	0.616	0.841	0.533
Hybrid RRF k=20 (Exp 1.2)	0.627	0.760	0.947	0.652
BM25+CSQE (Exp 013)	0.616	0.745	0.942	0.638
Dense+CSQE (Exp 013)	0.592	0.707	0.882	0.623
BM25-only-expanded CC (alpha=0.6)	0.709	0.830	0.972	0.727
BM25-only-expanded RRF (k=20)	0.714	0.836	0.973	0.736
Dense-only-expanded CC (alpha=0.4)	0.659	0.785	0.957	0.678
Dense-only-expanded RRF (k=20)	0.647	0.793	0.957	0.658
Both-expanded CC (alpha=0.5)	0.696	0.825	0.965	0.708
Both-expanded RRF (k=20)	0.694	0.829	0.966	0.704

Figure 4.11 v2 *System progression — HEADLINE*

Planned section: §4.9

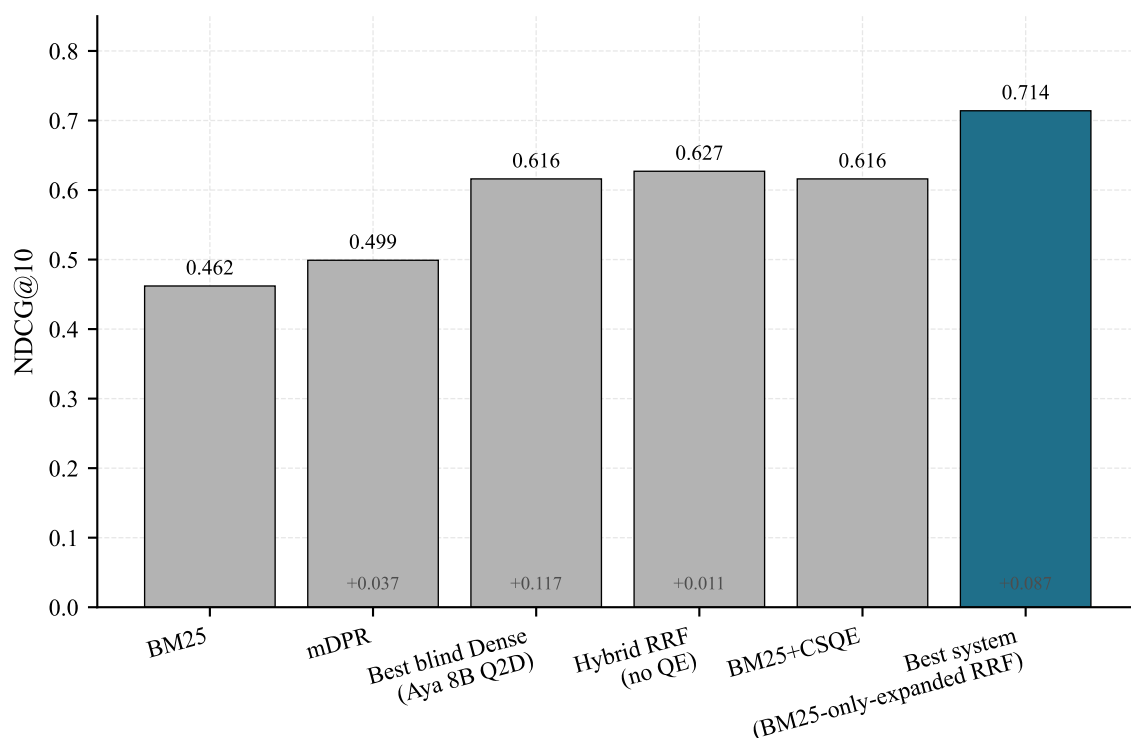


Figure 3.11: Progression of nDCG@10 from the BM25 baseline (0.462) through intermediate query-enhancement steps to the best system, BM25-only-expanded with RRF fusion, at 0.714. The annotation on each bar reports the gain over the best previous configuration.

Figure 4.12 *Per-query Δ nDCG@10 — CSQE+Hybrid vs Aya-blind*

Planned section: §4.10

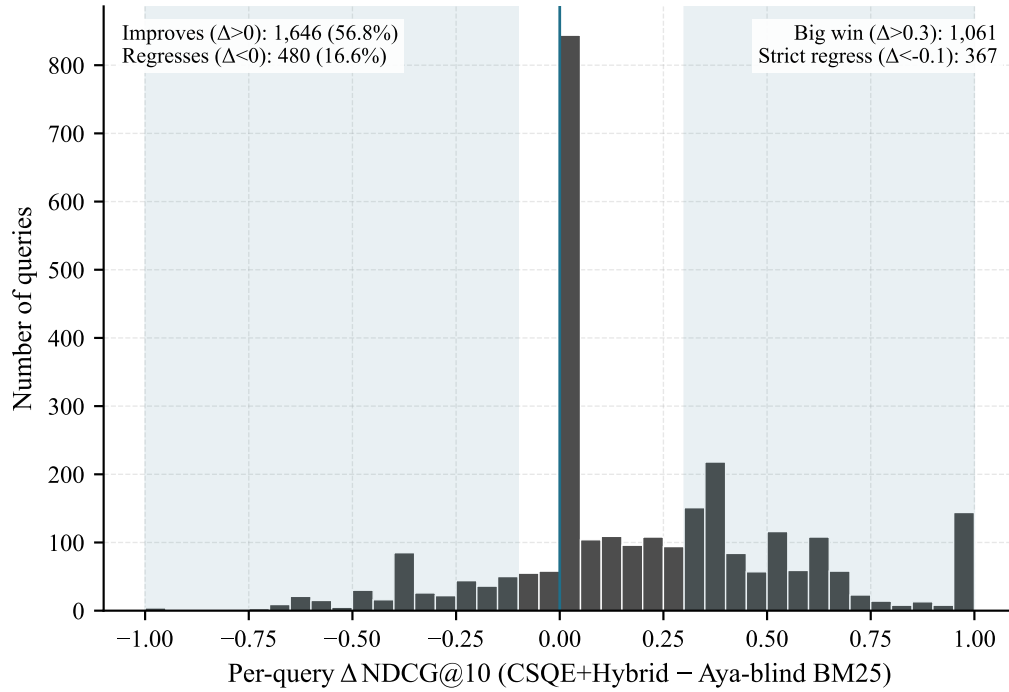


Figure 3.12: Distribution of the per-query difference between the CSQE+Hybrid system and the Aya-blind BM25 baseline. Shaded bands mark the Big Win region ($\Delta > 0.3$) and the Regression region ($\Delta < -0.1$); 56.8% of queries improve, 16.6% regress.

Figure 4.13 v2 *nDCG@10 by first-pass relevance (with sample sizes)*

Planned section: §4.10

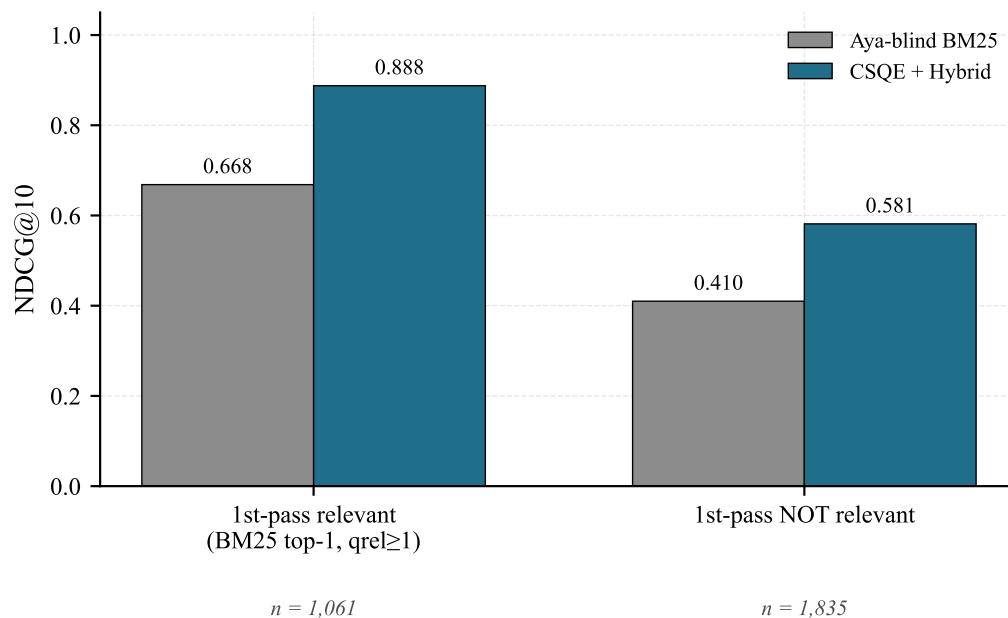


Figure 3.13: CSQE+Hybrid versus Aya-blind baseline conditioned on whether the BM25 first-pass top-1 document is relevant ($\text{qrel} \geq 1$). When the first-pass document is relevant ($n = 1,061$) CSQE+Hybrid reaches 0.888 nDCG@10; when it is not ($n = 1,835$) it reaches 0.581.

Figure 4.14 v2 $\Delta nDCG@10$ by query length, grouped blind vs CSQE

Planned section: §4.10

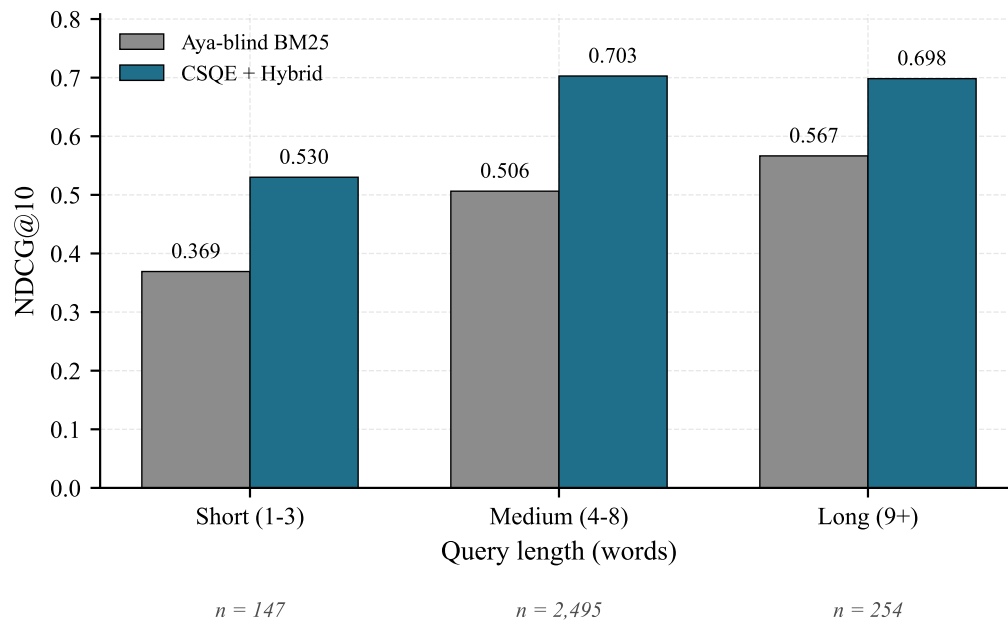


Figure 3.14: Per-bucket comparison of Aya-blind BM25 and CSQE+Hybrid using the 1–3 / 4–8 / 9+ word buckets. CSQE+Hybrid improves every bucket; short queries gain the most proportionally (+43.6%) and medium queries the most absolutely (+0.197).

Figure 4.15 v2 *Regression types breakdown (donut, $n = 367$)*

Planned section: §4.10

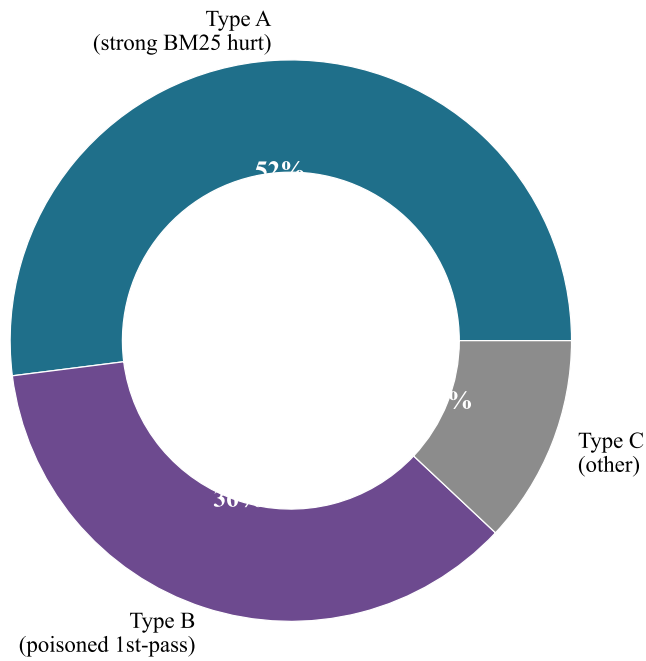


Figure 3.15: Type breakdown of the 367 queries that regress under CSQE+Hybrid ($\Delta < -0.1$). Type A (strong-BM25-hurt-by-expansion) accounts for 52%, Type B (poisoned first-pass) for 36%, and Type C (other) for 12%.